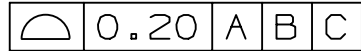
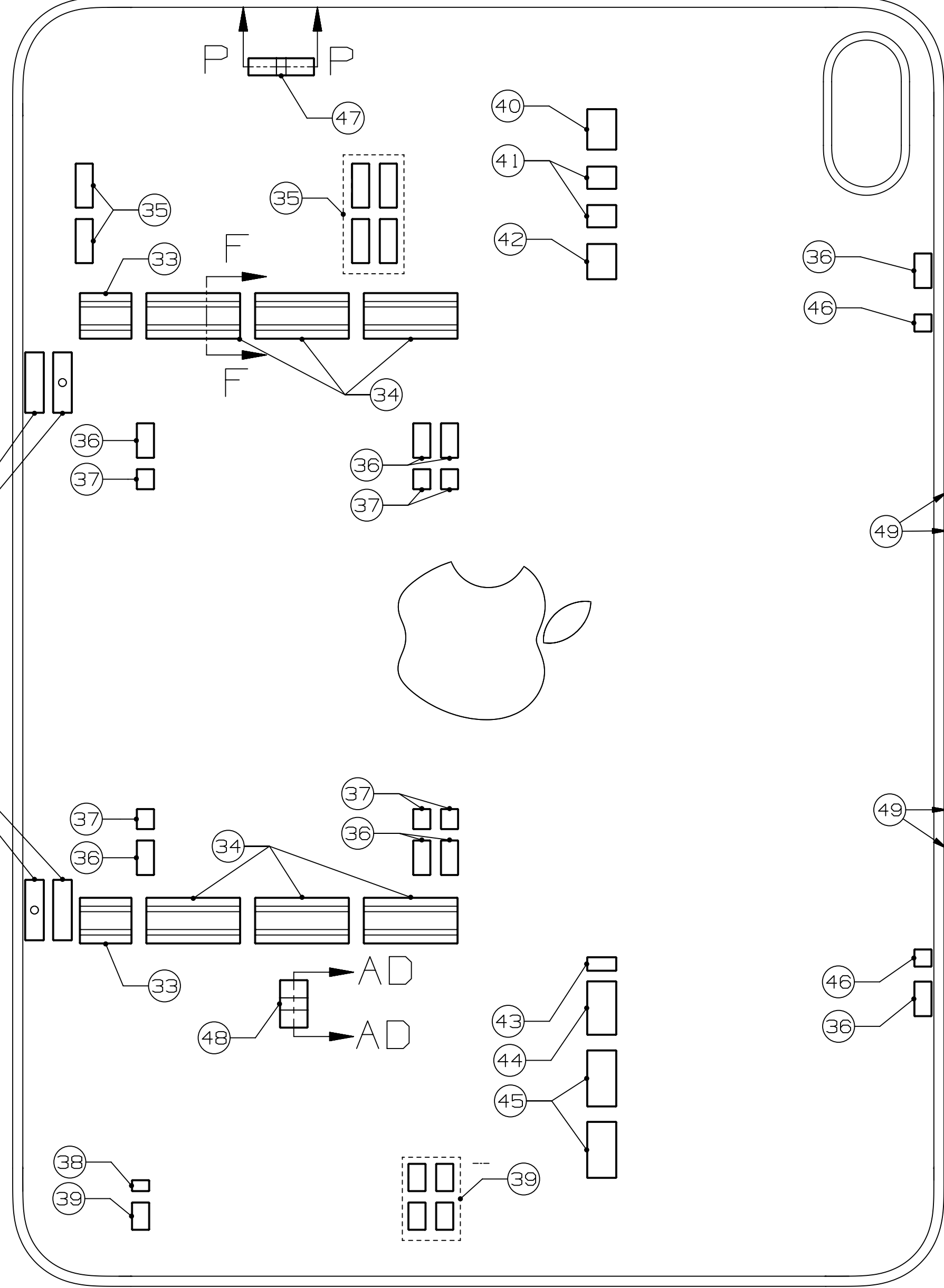
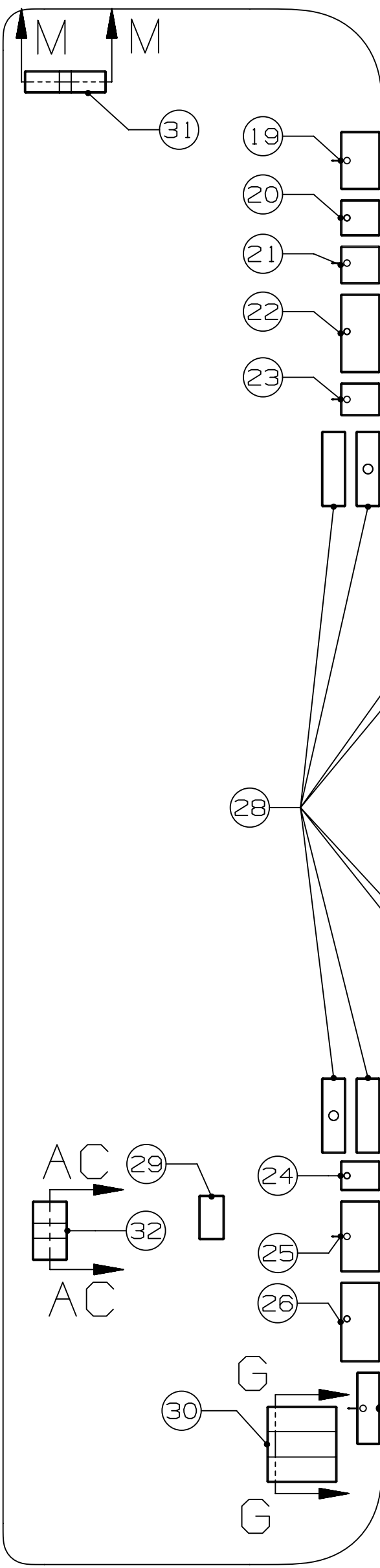
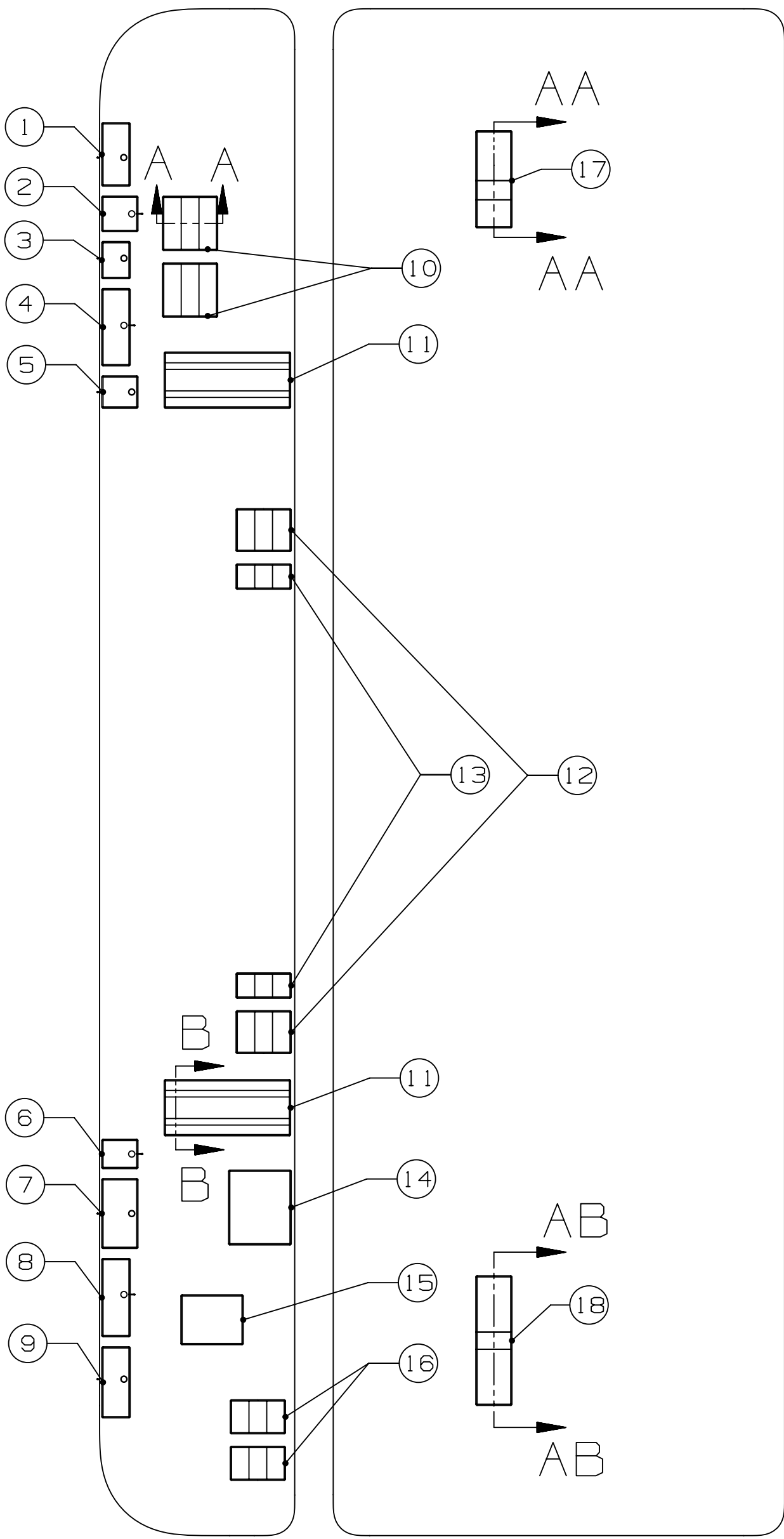
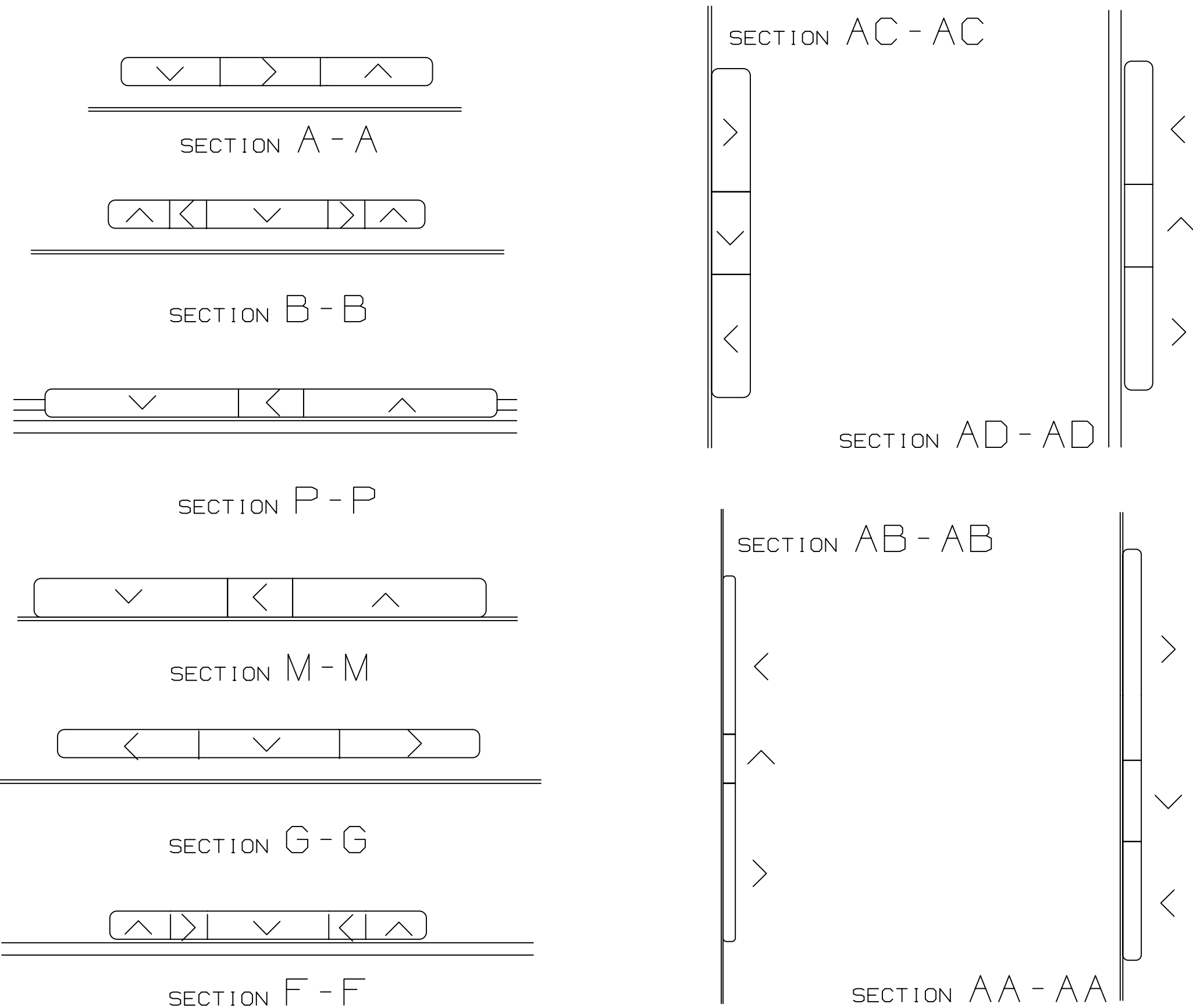
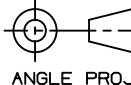


- NOTES:** (UNLESS OTHERWISE SPECIFIED)
- 1. THREE DIMENSIONAL CAD GEOMETRY AND BOMs PROVIDED BY APPLE INC. SHALL BE USED FOR CREATION OF THIS PART OR ASSEMBLY. DRAWING SHALL BE USED TO IDENTIFY CRITICAL TOLERANCES AND REFERENCE FEATURES.
 - 2. ALL DIMENSIONS AND TOLERANCES APPLY PER ASME Y14.5, 2009.
 - 3. DIMENSIONS DESIGNATED WITH THE  SYMBOL SHALL BE FIRST ARTICLE INSPECTION DIMENSIONS. THREE SAMPLE PARTS MUST BE FULLY INSPECTED. FAI REPORT MUST BE VALIDATED AND APPROVED BY APPLE INC. MQE.
 - 4. TOLERANCES DESIGNATED WITH THE  SYMBOL SHALL BE PRODUCED WITH STATISTICAL TOLERANCE PROCESS CONTROLS. IN ORDER TO USE THE TOLERANCE ON A FEATURE, THE STATISTICAL PROCESS (CP, CPK, ETC) OF THE PART MUST BE VALIDATED AND APPROVED BY APPLE INC. MQE.
 - 5. UNLESS OTHERWISE SPECIFIED, ALL SURFACES TO BE WITHIN A SURFACE PROFILE TOLERANCE OF  0.20 | A | B | C
 - 6. ALL TOOLING, FIXTURING AND OTHER UNIQUE ITEMS THAT ARE USED TO CREATE THIS PART ARE THE PROPERTY OF APPLE INC. AND SHALL BE PERMANENTLY MARKED IN ACCORDANCE WITH LATEST REVISION OF DOCUMENT "SECURITY RESPONSIBILITIES FOR APPLE SUPPLIERS"
 - 7. THE DESIGN OF ALL TOOLING OR FIXTURING REQUIRED FOR THE MANUFACTURING OR VERIFICATION OF THE PART SHALL BE APPROVED BY THE APPROPRIATE APPLE INC. ENGINEER PRIOR TO TOOL OR FIXTURE FABRICATION.
 - 8. ALL HOMOGENEOUS MATERIALS MUST COMPLY WITH THE APPLE REGULATED SUBSTANCES SPECIFICATION, 069-0135
 - ALL ADHESIVES, COATINGS AND PAINTS, PRINTING INKS, AND CLEANING AGENTS USED IN THE MANUFACTURING OF THIS PART MUST COMPLY WITH THE APPLE VOC SPECIFICATION, 099-22549
 - ALL MATERIALS WITH RECYCLED OR RENEWABLE CONTENT MUST COMPLY WITH THE APPLE RECYCLED & RENEWABLE MATERIAL SPECIFICATION, 099-15583
 - 9. PARTS TO BE FREE FROM MANUFACTURING RESIDUE OR DEBRIS.
 -  ALL ANGLED AND PRE MAGNETIZED MAGNETS ARE TO BE LASER MARKED DOUBLE SIDED. ALL SPECIAL MARK MAGNETS ARE TO BE LASER MARKED ON ONE SIDE.
 -  ALL MAGNET TOLERANCES X,Y,Z,RADII & POLARITY ANGLE FOUND ON SEPERATE SHEET
 -  MATERIAL: REFER TO MATERIAL TABLE IN SEPARATE SHEET
 -  99.5% YIELD CONTROL FOR MAGNETIZATION ANGLE
- ARROWS POINT TOWARDS NORTH
(IN-PLANE POLE ALWAYS POINTS FROM SOUTH TO NORTH IN OPEN 180 TOP VIEW)



METRIC		Apple Inc.	
DRAFTER	DATE	NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: (i) TO MAINTAIN THIS DOCUMENT IN CONFIDENCE (ii) NOT TO REPRODUCE OR COPY IT (iii) NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART (iv) ALL RIGHTS RESERVED	
APPLE PD	08/28/25		
DESIGNER	DATE		
APPLE PD	08/28/25		
DIMENSIONS ARE IN MILLIMETERS TOLERANCES		TITLE MAGNET, MAP, THINFPS, EXIT, R240	
DO NOT SCALE DRAWINGS		DRAWING NUMBER 099-53721	REV. 09
 THIRD ANGLE PROJECTION	SIZE D	SCALE NONE	SHT 1 OF 4

FAI 1 SPC A

FAI 2 SPC B

FAI 3 SPC C

FAI 5 SPC E 13

FAI 4 SPC D

ITEM	APN	NAME	X (MM)	Y (MM)	Z (MM)	GRADE	MAGNETIZATION ANGLE	QTY	MANUFACTURING PROCESS	MAGNETIZATION SEQUENCE	MAGNETIC MOMENT (uWb*cm)	LASER MARKING
1	160-05846	MAG,C,FH,HD,TRI,01,TW,THINFPS,R240	10.17	4.55	0.9	N42SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	4.65 TO 5.45	ANGLED MARKING
2	160-05847	MAG,C,FH,TRI,01,THINFPS,R240	5.69	5.8	0.9	N42SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	3.31 TO 3.89	ANGLED MARKING
3	160-05849	MAG,C,FH,HD,TRI,02,TW,THINFPS,240	5.96	4.55	0.9	N35SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	2.49 TO 2.93	ANGLED MARKING
4	160-05851	MAG,C,FH,HD,TRI,03,TW,THINFPS,240	12.38	4.55	0.9	N35SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	5.19 TO 6.09	ANGLED MARKING
5	160-05844	MAG,C,FH,TRI,02,THINFPS,R240	5.18	5.8	0.9	N42SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	3.02 TO 3.54	ANGLED MARKING
6	160-05852	MAG,C,CH,TRI,01,THINFPS,R240	4.7	5.8	0.9	N42SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	2.73 TO 3.21	ANGLED MARKING
7	160-05850	MAG,C,CH,TRI,02,THINFPS,R240	11.2	5.8	0.9	N42SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	6.54 TO 7.68	ANGLED MARKING
8	160-05848	MAG,C,CH,HD,TRI,01,TW,THINFPS,R240	12.56	4.55	0.9	N35SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	5.26 TO 6.18	ANGLED, NUMERICAL MARKING,
9	160-05845	MAG,C,CH,HD,TRI,02,TW,THINFPS,R240	11.41	4.55	0.9	N35SH	55° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	4.78 TO 5.62	ANGLED MARKING
10	160-05549	MAG,C,ED,01,R413	8.7	8.85	0.8	12H	REFER TO SECTION A-A AND 3D FOR POLARITY	2	RIGID BONDED	POST MAGNETIZED	3.04 TO 3.57	-
11	160-05550	MAG,C,SLIDER,OPT_1,R413	9	20.28	0.8	8	REFER TO SECTION B-B AND 3D FOR POLARITY	2	RIGID BONDED	POST MAGNETIZED	6.46 TO 7.58	-
12	160-05552	MAG,C,ED,02,R413	6.86	8.85	0.8	12H	REFER TO SECTION A-A AND 3D FOR POLARITY	2	RIGID BONDED	POST MAGNETIZED	2.39 TO 2.80	-
13	160-05553	MAG,C,ED,03,R413	4.06	8.85	0.8	12H	REFER TO SECTION A-A AND 3D FOR POLARITY	2	RIGID BONDED	POST MAGNETIZED	1.43 TO 1.67	-
14	160-05554	MAG,C,HES,R413	12	10.05	0.7	N35SH	NORMAL	1	SINTERED	POST MAGNETIZED	8.66 TO 10.17	-
15	160-05555	MAG,C,BAL,R413	8	10	0.8	N35SH	NORMAL	1	SINTERED	POST MAGNETIZED	6.66 TO 7.82	-
16	160-05556	MAG,C,ED,04,R413	5.46	8.85	0.8	12H	REFER TO SECTION A-A AND 3D FOR POLARITY	2	RIGID BONDED	POST MAGNETIZED	1.93 TO 2.26	-
17	160-05842	MAG,B,STICK,THINFPS,01,R240	15.58	5.8	0.7	12H	REFER TO SECTION K-K AND 3D FOR POLARITY	1	RIGID BONDED	POST MAGNETIZED	3.13 TO 3.67	-
18	160-05843	MAG,B,STICK,THINFPS,02,R240	20.79	5.8	0.7	12H	REFER TO SECTION L-L AND 3D FOR POLARITY	1	RIGID BONDED	POST MAGNETIZED	4.17 TO 4.89	-
19	160-05861	MAG,A,FH,TRI,01,TW,THINFPS,R240	9.15	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	7.71 TO 9.05	ANGLED MARKING
20	160-05857	MAG,A,FH,TRI,02,THINFPS,R240	5.69	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	4.78 TO 5.62	ANGLED, NUMERICAL MARKING,
21	160-05858	MAG,A,FH,TRI,03,TW,THINFPS,R240	5.95	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	5.00 TO 5.88	ANGLED MARKING,
22	160-05859	MAG,A,FH,TRI,04,TW,THINFPS,R240	12.38	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	10.45 TO 12.27	ANGLED MARKING
23	160-05860	MAG,A,FH,TRI,05,THINFPS,R240	5.18	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	4.35 TO 5.11	ANGLED MARKING
24	160-05855	MAG,A,CH,TRI,01,THINFPS,R240	4.7	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	3.95 TO 4.63	ANGLED MARKING
25	160-05856	MAG,A,CH,TRI,02,THINFPS,R240	11.2	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	9.45 TO 11.09	ANGLED MARKING
26	160-05853	MAG,A,CH,TRI,03,TW,THINFPS,R240	12.56	6.2	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	10.60 TO 12.44	ANGLED, NUMERICAL MARKING,
27	160-05854	MAG,A,CH,TRI,04,TW,THINFPS,R240	11.41	3.67	1.15	N45SH	160° OFF Y IN YZ PLANE	1	SINTERED	PRE AND POST MAGNETIZED	5.65 TO 6.63	ANGLED MARKING
28	160-05841	MAG,A,ACRUSH,THINFPS,01,R240	11.85	3.8	0.8	N35SH	NORMAL	8	SINTERED	PRE AND POST MAGNETIZED	3.62 TO 4.26	POLARITY MARKING
29	160-05570	MAG,A,BAL,R413	7	4	0.8	N35SH	NORMAL	1	SINTERED	POST MAGNETIZED	2.32 TO 2.72	-
30	160-05571	MAG,A,HES,BONDED,R413	12	11	0.8	12H	REFER TO SECTION G-G AND 3D FOR POLARITY	1	RIGID BONDED	POST MAGNETIZED	5.15 TO 6.04	-
31	160-05748	MAG,A,STICK,01,R240	3.5	12.85	1.1	12H	REFER TO SECTION M-M AND 3D FOR POLARITY	1	RIGID BONDED	POST MAGNETIZED	3.02 TO 2.58	-
32	160-05744	MAG,A,STICK,02,R240	9.35	5.5	1.1	12H	REFER TO SECTION N-N AND 3D FOR POLARITY	1	RIGID BONDED	POST MAGNETIZED	3.43 TO 2.93	-
33	160-06672	MAG,BACK,RAIL,01,R240	9	10.15	0.8	8	REFER TO SECTION F-F AND 3D FOR POLARITY	2	RIGID BONDED	POST MAGNETIZED	3.18 TO 3.74	-
34	160-06671	MAG,BACK,RAIL,02,R240	9	18.28	0.8	8	REFER TO SECTION F-F AND 3D FOR POLARITY	6	RIGID BONDED	POST MAGNETIZED	5.68 TO 6.68	-
35	160-05583	MAG,BACK,ED,01,R413	8.7	3.5	0.8	N35SH	NORMAL	6	SINTERED	POST MAGNETIZED	2.49 TO 2.92	-
36	160-05584	MAG,BACK,ED,02,R413	6.86	3.5	0.8	N35SH	NORMAL	8	SINTERED	POST MAGNETIZED	1.98 TO 2.33	-
37	160-05585	MAG,BACK,ED,03,R413	4.06	3.5	0.8	N35SH	NORMAL	6	SINTERED	POST MAGNETIZED	1.17 TO 1.37	-
38	160-05586	MAG,BACK,ED,04,R413	2.3	3.5	0.8	N35SH	NORMAL	1	SINTERED	POST MAGNETIZED	0.65 TO 0.76	-
39	160-05587	MAG,BACK,ED,05,R413	5.46	3.5	0.8	N35SH	NORMAL	5	SINTERED	POST MAGNETIZED	1.57 TO 1.85	-
40	160-05589	MAG,BACK,PEEK,01,TW,R413	8	5.8	0.8	N45SH	NORMAL	1	SINTERED	POST MAGNETIZED	4.24 TO 4.97	-
41	160-05590	MAG,BACK,PEEK,02,R413	4.5	5.8	0.8	N45SH	NORMAL	2	SINTERED	POST MAGNETIZED	2.38 TO 2.80	-
42	160-05750	MAG,BACK,PEEK,03,TW,R240	7	5.8	0.8	N45SH	NORMAL	1	SINTERED	POST MAGNETIZED	3.68 TO 4.32	-
43	160-05592	MAG,BACK,PEEK,04,R413	2.8	5.8	0.8	N45SH	NORMAL	1	SINTERED	POST MAGNETIZED	1.47 TO 1.73	-
44	160-05593	MAG,BACK,PEEK,05,R413	10.5	5.8	0.8	N45SH	NORMAL	1	SINTERED	POST MAGNETIZED	5.59 TO 6.56	-
45	160-05752	MAG,BACK,PEEK,06,TW,R240	11	5.8	0.8	N45SH	NORMAL	2	SINTERED	POST MAGNETIZED	6.8 TO 5.8	-
46	160-05557	MAG,B,BACK,MULTI,01,R413	3.5	3.5	0.8	N35SH	NORMAL	2	SINTERED	POST MAGNETIZED	0.99 TO 1.17	-
47	160-05755	MAG,BACK,ASTICK,01,R240	3.5	12.85	0.8	12H	REFER TO SECTION P-P AND 3D FOR POLARITY	1	RIGID BONDED	POST MAGNETIZED	1.84 TO 2.16	-
48	160-05756	MAG,BACK,ASTICK,02,R240	9.35	5.5	0.8	12H	REFER TO SECTION R-R AND 3D FOR POLARITY	1	RIGID BONDED	POST MAGNETIZED	2.09 TO 2.45	-
49	160-06150	MAG,PENCIL,SHELL,0P62,01,R240	3.71	0.62	3.9	N50SH	NORMAL	4	SINTERED	POST MAGNETIZED	1.02 TO 1.18	LINEAR MARKING
50	160-06149	MAG,PENCIL,SHELL,0P62,02,R240	3.71	0.62	3.9	N50M	NORMAL	4	SINTERED	PRE AND POST MAGNETIZED	0.99 TO 1.17	-

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
(iii) TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
(iiii) NOT TO REPRODUCE OR COPY IT
(v) NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
(vi) ALL RIGHTS RESERVED

Apple Inc.

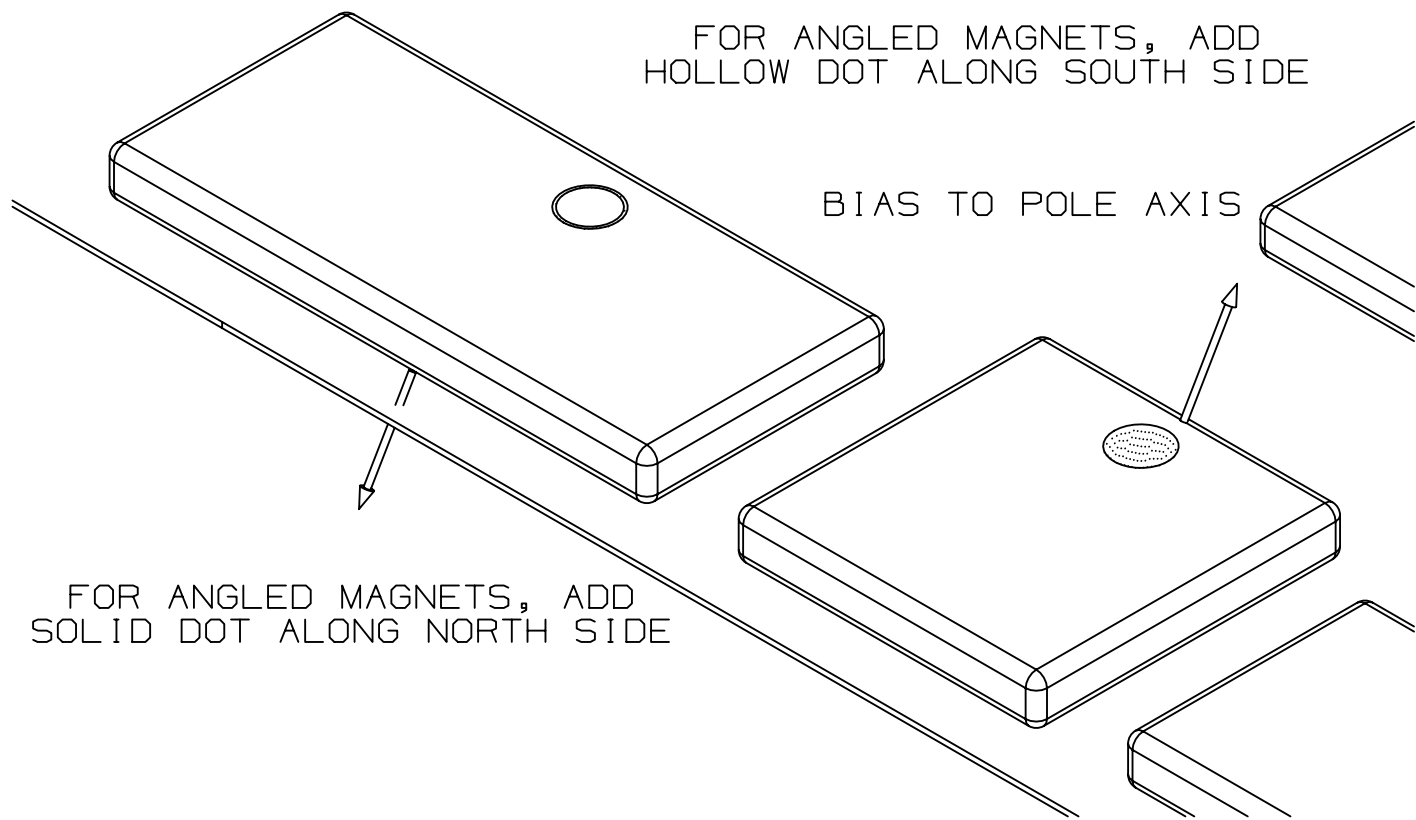
SIZE	DRAWING NUMBER	REV.
D	099-53721	09
SCALE:	NONE	SHT 2 OF 4

MATERIAL SPEC¹²

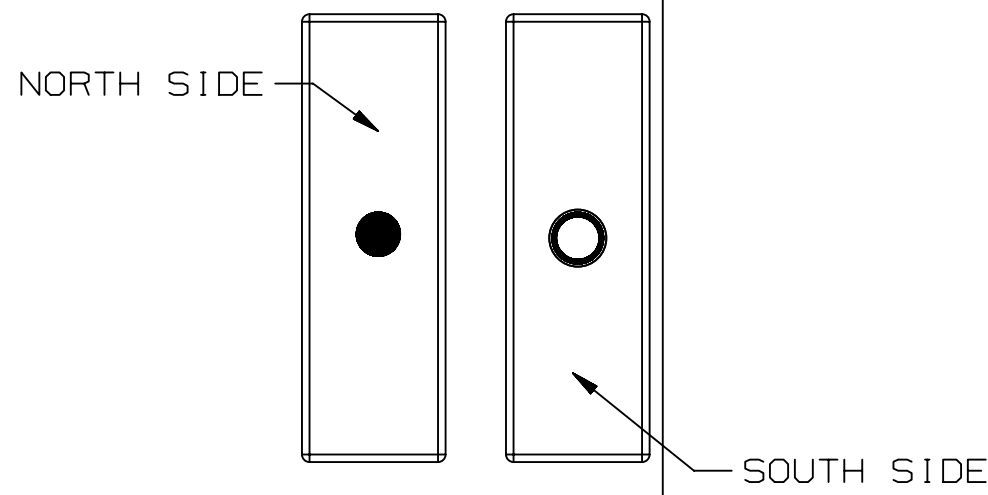
MATERIAL GRADE	GRADE INFO	ERS	Br (kGs)	Hcb (KOe)	Hc _J (KOe)	(BH)max (MGOe)	COATING
B	B-100RC (REE)	080-02981	6.0-6.8	4.5-5.5	8.0-10.0	7.5-9.0	10-20 μm BLACK EPOXY
8H	8H-100RC (REE)	080-02981	6.0-6.6	5.0-6.0	15.0-17.0	8.0-9.0	10-20 μm BLACK EPOXY
10H	10H-100RC (REE)	080-02981	6.7-7.5	5.5-6.5	9.0-11.0	9.5-11.0	10-20 μm BLACK EPOXY
12H	12H-100RC (REE)	080-02981	7.2-8.0	5.5-6.5	9.0-10.0	11.0-12.0	10-20 μm BLACK EPOXY
N35SH	N35SH-100RC (REE)	080-02981	11.7 - 12.2	-	20 MIN	33-36	6-14 μm VERY BRIGHT ZINC
N42SH	N42SH-100RC (REE)	080-02981	12.8 - 13.2	-	20 MIN	40-43	6-14 μm VERY BRIGHT ZINC
N45SH	N45SH-100RC (REE)	080-02981	13.2-13.8	-	20 MIN	42.98-45.99	6-14 μm VERY BRIGHT ZINC
N50M	N50M- 100RC (REE)	080-02981	13.9 - 14.4	-	14 MIN	47-51	7-19 μm NICUNI
N50SH	N50SH- 100RC (REE)	080-02981	13.9 - 14.4	-	20 MIN	47-51	7-19 μm NICUNI
STEEL 1010	SPCD STEEL, 0.10% MAX CARBON CONTENT	-	-	-	-	-	6-14 μm VERY BRIGHT ZINC

NOTE: ANNEAL STEEL 1010 SHUNTS PRIOR TO PLATING, USING FOLLOWING PARAMETERS: HEAT TO 900-980C FOLLOWED BY SLOW COOLING IN FURNCE. FURNACE ENVIRONMENT: CREATE VACUUM IN THE FURNACE AND FILL WITH DRY HYDROGEN OR DISSOCIATED AMMONIA.

LASER MARKING¹⁰

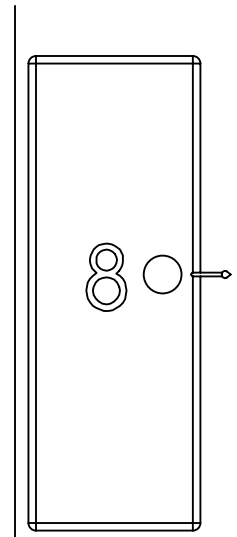


FOR "POLARITY MARKING", ADD SOLID DOT ON NORTH SIDE, ADD HOLLOW DOT ON SOUTH SIDE



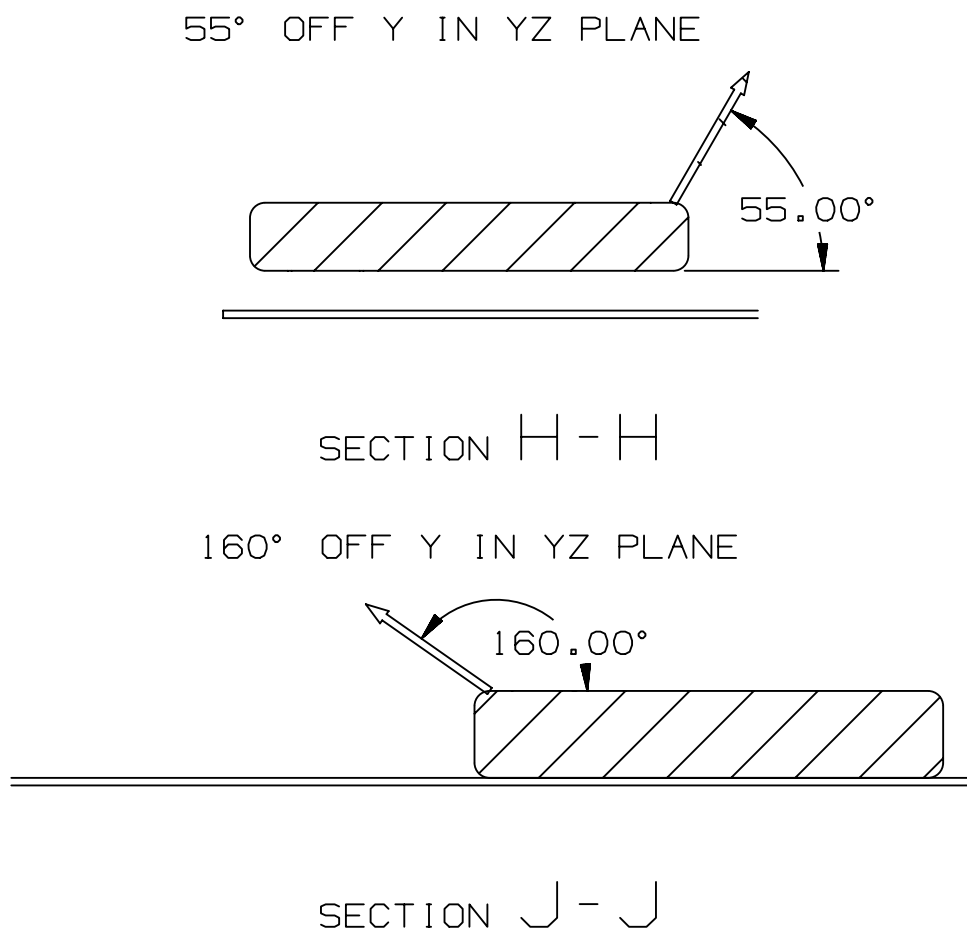
MAGNET NUMBER	MARKING
29	POLARITY MARKING

FOR "NUMERICAL MARKING", ADD MAGNET NUMBER FROM MAGNET LIST ON NORTH, SOUTH SIDE

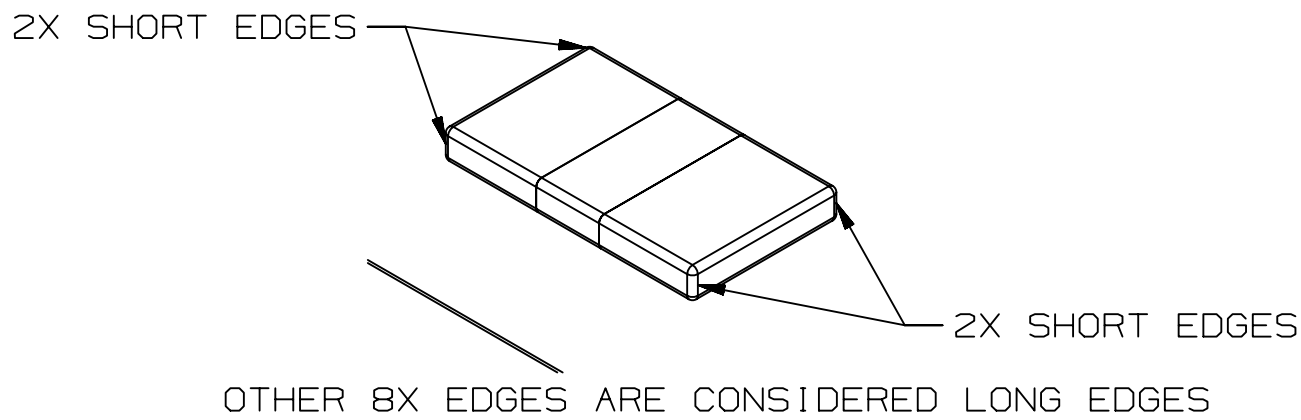


MAGNET NUMBER	APN	MARKING
B	160-05848	ANGLED, LINEAR MARKING

POLARITY ANGLE¹¹



MAGNET EDGE DIAGRAM¹¹



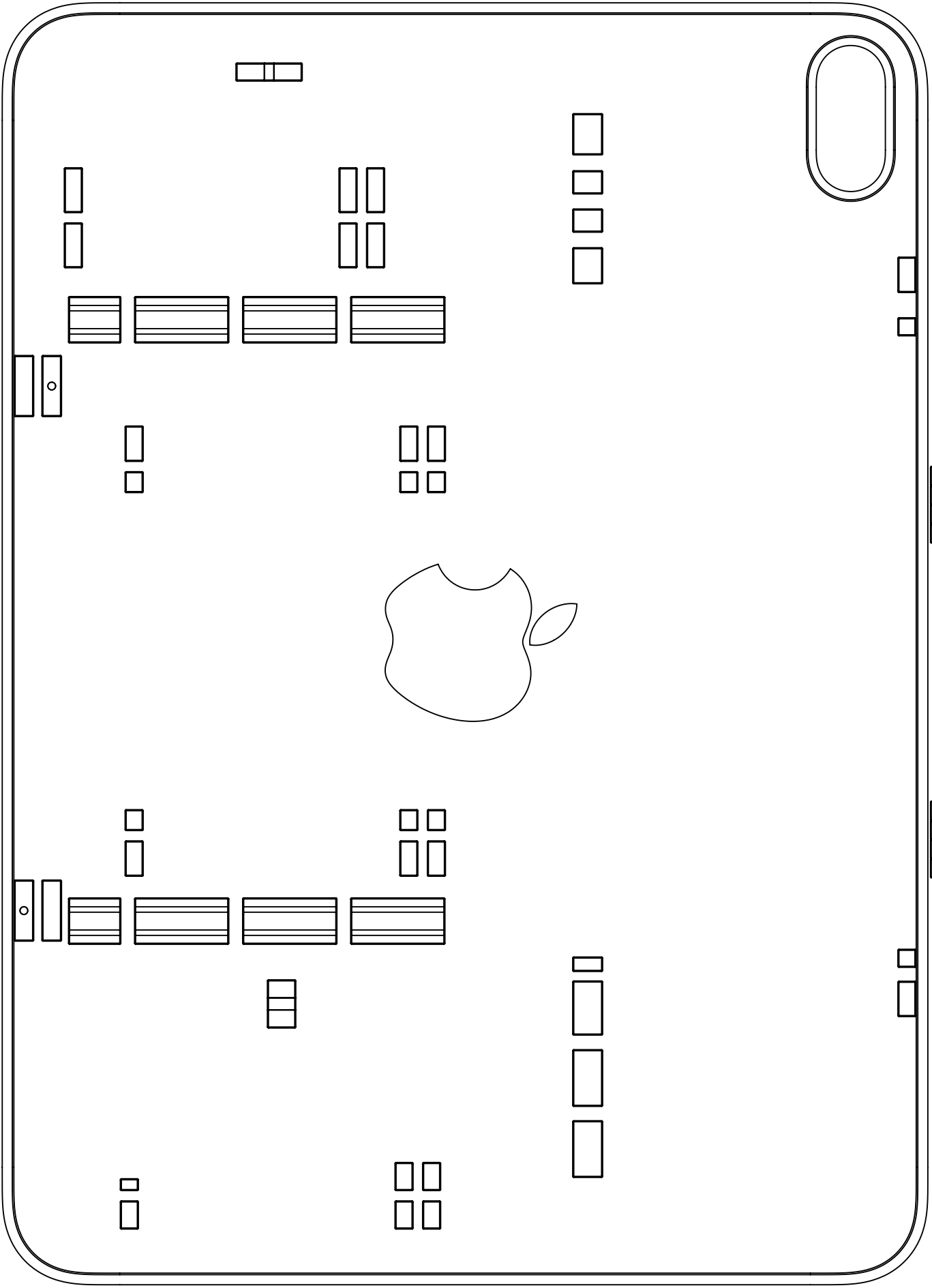
DIMENSIONAL TOLERANCES¹¹

DIMENSION	SINTERED MAGNET TOLERANCE	BONDED MAGNET TOLERANCE	SHUNT TOLERANCE
X	±0.05mm	±0.05mm	±0.05mm
Y	±0.05mm	±0.05mm	±0.05mm
Z	±0.04mm	±0.04mm	±0.03mm
RADII - 4X SHORT EDGE	0.20 +0.10/-0.08mm	0.25 ± 0.10mm	0.30mm MAX
RADII - 8X LONG EDGE	0.20mm MAX	0.20mm MAX	-
POLARITY ANGLE	±7°	-	-

FAL 6

FAL 7

13



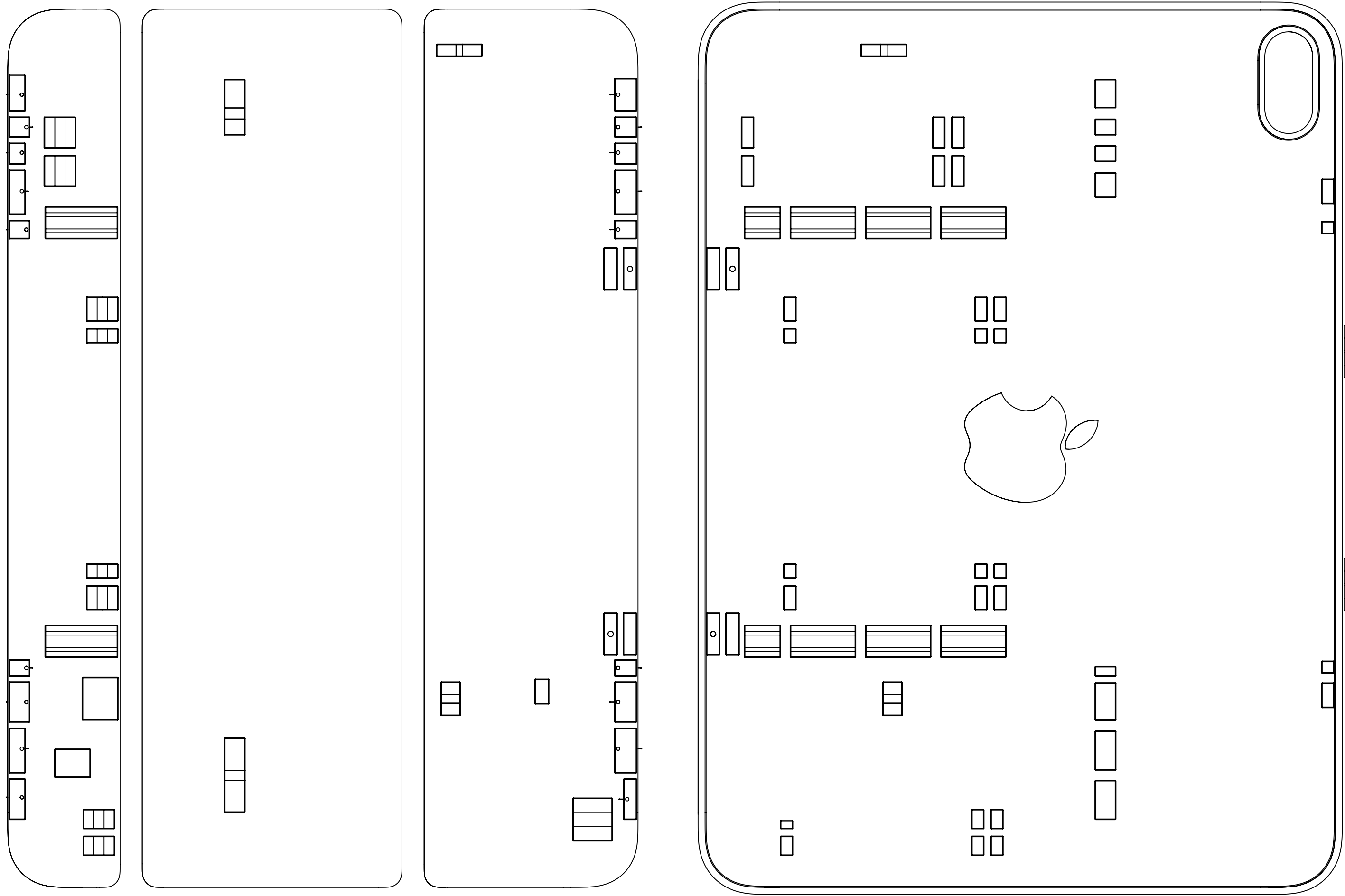
NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
(1) TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
(2) NOT TO REPRODUCE OR COPY IT
(3) NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
(4) ALL RIGHTS RESERVED

Apple Inc.

SIZE	DRAWING NUMBER	REV.
D	099-53721	09
SCALE	NONE	SHT 3 OF 4

SIDEWALL MAGNETS



AJ

AK

AJ

POLARITY
ALL ARROWS FACE NORTH

MAGNET ID

50

49

50

49

49

50

49

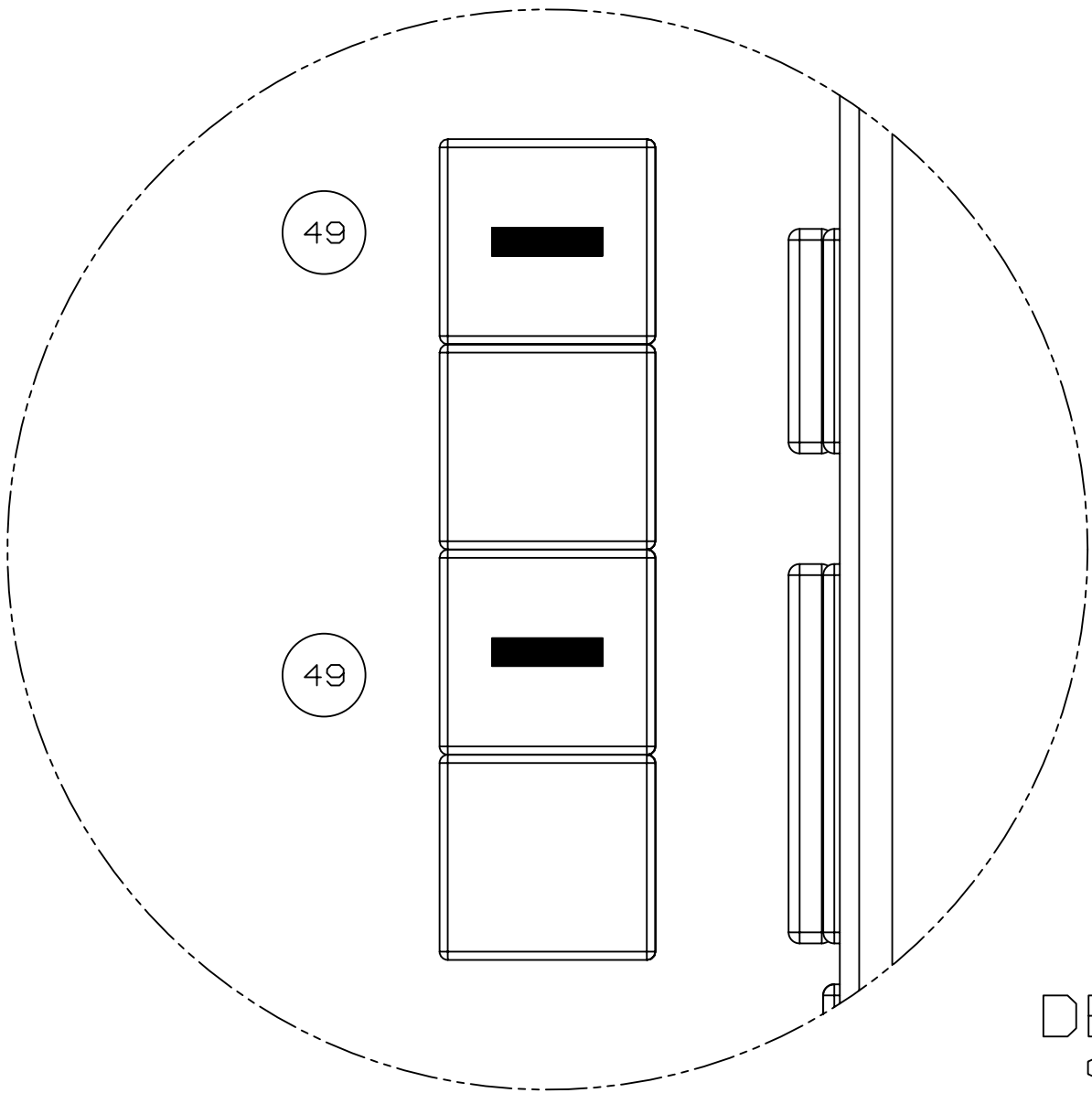
50

SECTION AJ-AJ

LINEAR MARKING

ONLY IF REQUESTED
IN MAGNET MAP

ADD LINE ACROSS
MAGNETS POLARIZED
THROUGH PLANE



DETAIL AK
SCALE 8:1